SDS 357: Case Studies in Data Science

Nashville SOPP:

Effect of Neighborhood Wealth on Stop Outcomes

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1. Introduction

1.1. Project Background

Our team uses the Stanford Open Policing Project (SOPP) dataset to examine traffic stop outcomes in Nashville, Tennessee. Nashville is widely known as a socioeconomically and racially diverse city, thus making it one of the most ideal settings to study how differing wealth levels across different parts of Nashville impact policing outcomes. To capture the variations in wealth across different areas of Nashville, we implemented data from the Nashville Housing dataset (NH) and constructed a wealth index based on Nashville housing sale prices.

Having an understanding of how traffic stop outcomes vary across Nashville is vital for public safety and evaluating equity and transparency in policing practices. Differences in stop outcomes, such as warnings, citations, and arrests, may reflect underlying socioeconomic disparities and possible discriminatory practices. By combining both policing and housing data, this project aims to investigate whether or not and how these wealth disparities are associated with measurable differences in traffic stop outcomes. By adding another layer that takes into account whether demographic factors such as race, sex, and age also associate with stop outcomes.

1.2. Project Objectives

Our main project objective is to investigate the relationship between stop outcomes and neighborhood wealth. We do this by quantifying neighborhood wealth into a wealth index variable and accounting for an individual's demographic characteristics (sex, age, race) with interaction terms. By controlling for demographics, we can compare the distribution of stop outcomes across neighborhoods with varying wealth indices. This allows us to better determine if the overall wealth index of a neighborhood affects the stop outcome an individual receives. To analyze this relationship, we chose a Multinomial Logistic Regression (MLR) model and evaluated our results by interpreting the model coefficients and predicted probabilities.

1.3. Data Science Pipeline Overview

To achieve our project objectives, we began by downloading Nashville policing data from the Stanford Open Policing Project and housing sale price data from the Nashville Housing Data portal. From there, we performed data cleaning and preprocessing, including filtering for rows where the year columns aligned across both the housing and policing datasets. In the policing dataset, we filtered for rows that included our key variables: latitude, longitude, subject sex, subject race, and stop outcome. Then we dropped rows with NA values in our future key variable columns. For our categorical variables of race, sex, and stop outcome, we encoded them for modeling purposes.

For our supplemental housing data, we dropped rows with NA values in the property address column as a preliminary step for geocoding. Then we used OpenStreetMap's API to geocode each address in the housing dataset, assigning exact latitude and longitude to each row. We filtered the housing data to only include the sale date, address, latitude, longitude, sale price, and address columns; the columns that are relevant in constructing the wealth index. We used housing prices to construct our wealth index variable and combined it with our policing dataset using coordinates and dates to create our main data frame.

Continuous variables such as age and wealth are centered to improve interpretability and reduce multicollinearity when interaction terms are introduced. We also construct a cluster identifier based on year and geographic location to account for potential spatial and temporal dependence in the data.

Next, we define our model specification using an MLR statistical model because our research question is inferential in nature. The model includes the wealth index, age, race, and sex as predictors, along with interaction terms between wealth and demographic variables to capture heterogeneity in effects. Prior to estimation, we conduct a series of diagnostic and assumption checks, including assessments of missingness, class balance, multicollinearity (via variance inflation factors), linearity (using Box–Tidwell tests), and sensitivity to the independence of irrelevant alternatives (IIA) assumption. Where necessary, we adjust the model specification—for example, by incorporating nonlinear terms (splines) for age or wealth—to better fit the data.

For model estimation, we split the data temporally. We fit the MLR model to obtain interpretable coefficients, standard errors, and statistical inference. Cluster-robust standard errors are used in the inference model to account for within-location dependence.

Finally, we generated post-estimation analyses and visualizations to interpret the model results. These include predicted probabilities of each traffic stop outcome at each level of the wealth index. We then used plots to illustrate how these relationships vary by race, sex, and age. We also computed marginal effects to quantify how changes in neighborhood wealth impact the probability of different outcomes for specific demographic groups.

2. Data and Data Exploration

2.1. Data Description

The SOPP dataset is a publicly available collection of traffic stop records from law enforcement agencies across the United States. The data is collected from official police department records and includes detailed information on individual traffic stops. This includes date and location of the stop, driver demographics (race, age, sex), and the outcome of the stop (warning, citation, arrest).

The NH dataset contains information on property transactions, including sale prices and geographic coordinates. Using this data, we construct a wealth index that serves as a proxy for neighborhood wealth. This allowed us to analyze how policing outcomes vary across different socioeconomic environments.

Prior to data cleaning and data wrangling, the policing dataset had over 3.8 million rows, and the housing dataset had over 56,000 rows. Afterwards, our final dataframe includes the key variables of latitude, longitude, date, year, subject age, subject race encoded, subject sex encoded, outcome encoded, and wealth index, with over 1.2 million rows.

2.2. Data Wrangling

Before conducting exploratory data analysis and modeling, we performed extensive data preprocessing and cleaning on both the SOPP and NH datasets. For the policing dataset, we first filtered observations to include only traffic stops that occurred between 2013 and 2016, to align temporally with the housing data. We standardized geographic information by rounding latitude and longitude values to six decimal places, which reflects standard geographic precision. We then removed columns that were not relevant to our analysis, retaining only variables necessary for modeling traffic stop outcomes and demographic characteristics. To address missing data, we dropped observations with missing values in key variables. We then encoded categorical variables such as race, sex, and outcome into numeric and binary formats to make them suitable for regression modeling. We subsequently removed the original categorical columns to avoid redundancy and multicollinearity.

A similar preprocessing pipeline was applied to the supplemental NH dataset. We filtered the data to remove observations with missing property address values to enable geocoding. Using OpenStreetMap's Nominatim API, we converted each property address into latitude and longitude coordinates, allowing us to spatially align the housing data with the traffic stop data. After geocoding, we removed the original address column since geographic coordinates provide more precise location information, dropped non-essential variables, and rounded latitude and longitude

values to six decimal places for consistency. We also removed any remaining observations with missing geographic coordinates to ensure all records could be used in spatial analysis.

To construct the wealth index, we used the latitude and longitude from both datasets. We found every housing sale within 0.5 miles, 1 mile, or the nearest 30 for comparison purposes. We then took the median of these values to construct an indicator of wealth for each stop.

The final cleaned dataset comprises over 1.3 million traffic stop observations and approximately 24,000 housing records, yielding a comprehensive, geographically aligned, and model-ready dataset suitable for statistical inference.

2.3. Exploratory Data Analysis (EDA)

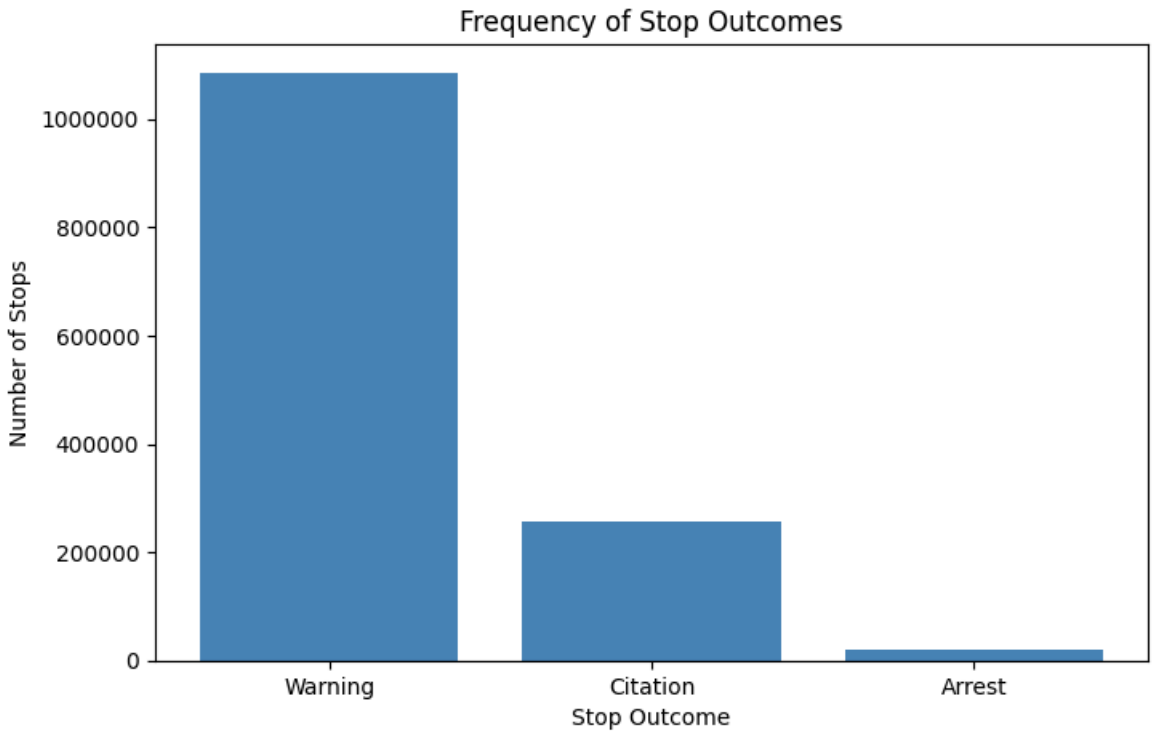

Figure 1. Distribution of Traffic Stop Outcomes

Figure 1 shows the distribution of traffic stop outcomes in the dataset. The majority of stops result in warnings, while citations occur much less frequently, and arrests are relatively rare. This indicates a strong class imbalance in the outcome variable. This imbalance is important for two reasons. First, it means that accuracy alone is not a reliable measure of model performance, since a model that always predicts “warning” would appear highly accurate without providing meaningful insight into citations or arrests. Second, the imbalance helps explain why the model has greater difficulty predicting arrests and citations than warnings. Because of this, our analysis focuses primarily on interpreting relationships between variables and stop outcomes rather than maximizing predictive accuracy.

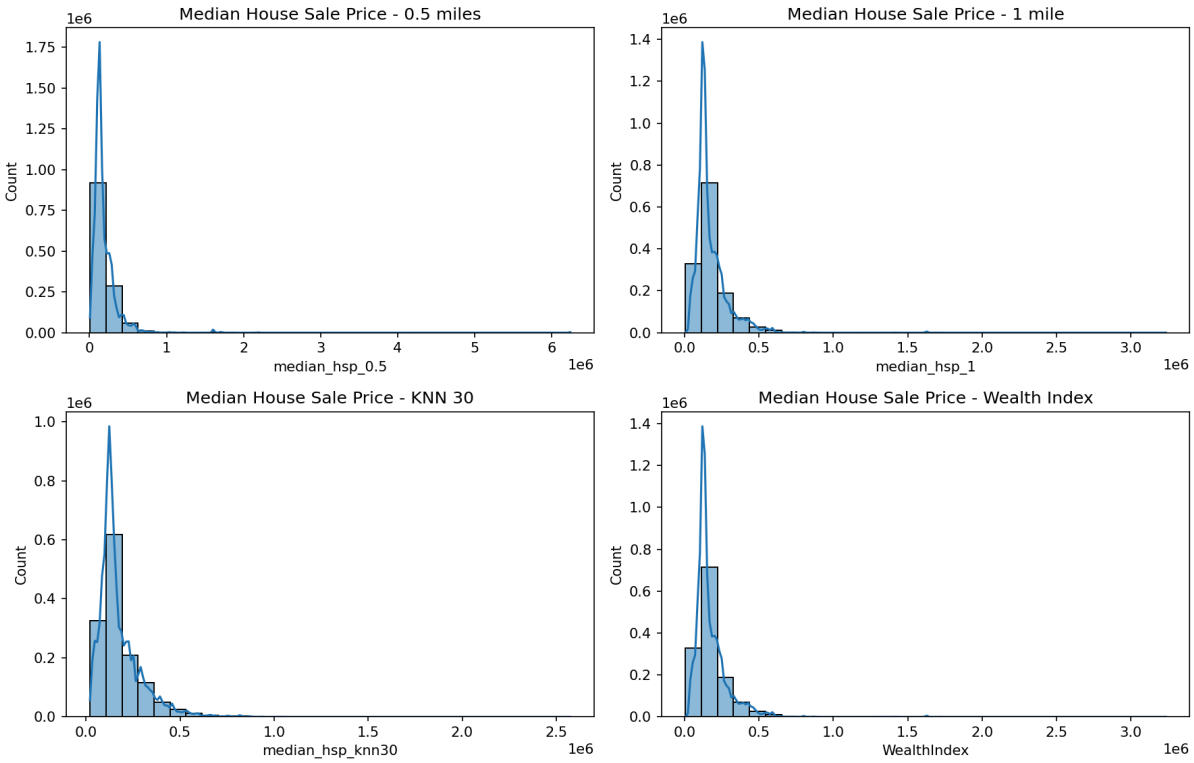

Figure 2. Comparison of Median House Sale Price Distributions Across Neighborhood Definitions

Figure 2 compares the distribution of median house sale prices under different neighborhood definitions used to construct the wealth index. The 0.5-mile radius measure appears more variable and sensitive to extreme values, likely because it is based on fewer nearby housing observations. The KNN-based measure produces a smoother distribution but is less geographically interpretable. The 1-mile radius measure provides a balance between stability and geographic interpretability, producing a distribution that is less sensitive to extreme values while still reflecting local neighborhood conditions. For this reason, the 1-mile radius median house price was selected to construct the wealth index variable used in the regression analysis

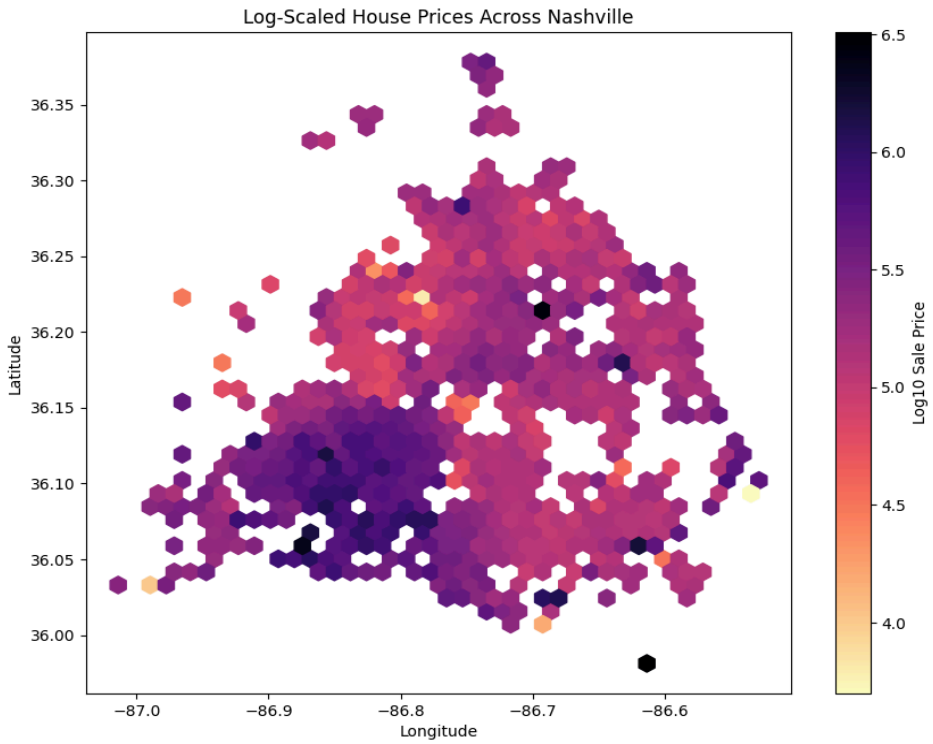

Figure 3. Spatial Distribution of Log-Scaled Median House Prices Across Nashville

Figure 3 shows that housing prices, and therefore neighborhood wealth, vary substantially across geographic areas of Nashville. Some regions exhibit consistently higher housing prices, while others show lower price levels, indicating clear spatial clustering in neighborhood wealth. This spatial structure is important because traffic stops occurring in nearby locations are likely influenced by similar neighborhood characteristics and policing environments. This pattern supports using neighborhood-level wealth as a key explanatory variable and a temporally separated train-test split to reduce the effects of spatial dependence and geographic leakage.

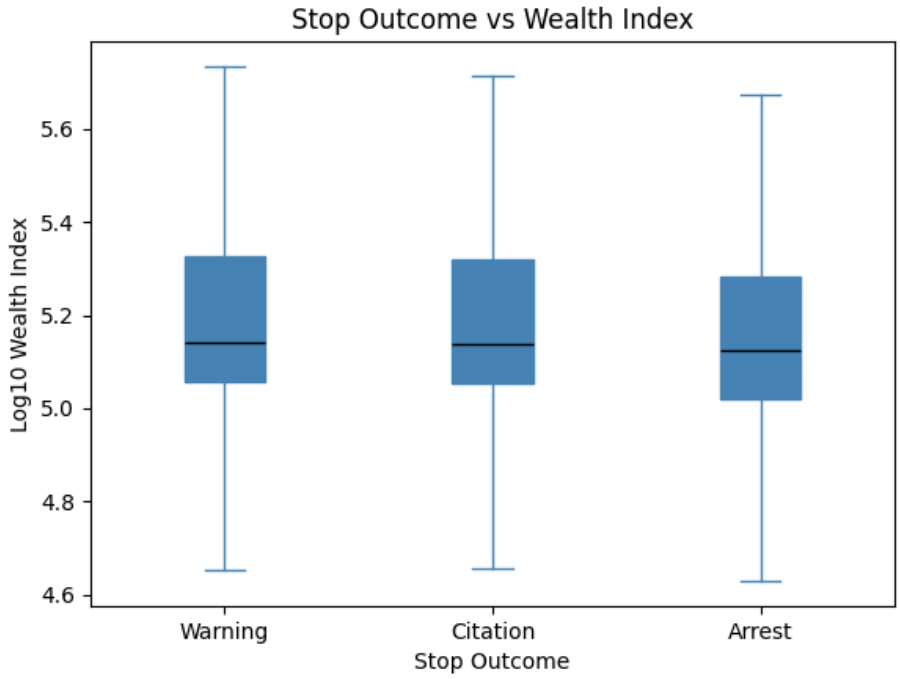

Figure 4. Stop Outcome vs. Wealth Index

Figure 4 shows the distribution of neighborhood wealth across different stop outcomes. The median wealth index appears slightly higher for warnings compared to citations and arrests, although the distributions overlap substantially. This suggests that stops occurring in wealthier neighborhoods may be somewhat more likely to result in warnings rather than more severe outcomes such as citations or arrests. However, the large overlap between the distributions indicates that neighborhood wealth alone does not determine stop outcomes and that other factors, such as demographic characteristics, must also be considered. This observation supports the use of a multivariable regression model to evaluate the relationship between neighborhood wealth and stop outcomes while controlling for race, age, and sex.

3. Methods and Models

3.1. Technical Approach

Our research question is inferential rather than purely predictive: does neighborhood wealth have an association with traffic stop outcomes, and does that association differ by race, sex, and age? Because the goal is to estimate and interpret relationships rather than simply maximize classification accuracy, we adopted an inferential statistical modeling strategy rather than a black-box machine learning approach.

The project is centered on understanding the direction, magnitude, and heterogeneity of associations between neighborhood wealth and stop outcomes. That requires a model whose parameters can be interpreted directly. Naturally, through interaction terms in a regression framework, we can examine whether the relationship between wealth and stop outcome varies across demographic groups. The underlying question is policy-oriented. We are not only interested in whether stop outcomes can be predicted, but also in whether neighborhood context is systematically associated with those outcomes after adjusting for subject characteristics.

For these reasons, our primary model is a Multinomial Logistic Regression. This model is appropriate because the response variable, stop outcome, is categorical with three exclusive classes: warning, citation, and arrest. These outcomes are not naturally continuous, and while one could argue that there is some severity ordering, the decision process that generates them is not guaranteed to satisfy the assumptions of an ordered model. An MLR allows each non-reference outcome to have its own log-odds equation relative to a baseline category, which provides flexibility without imposing an unnecessarily restrictive structure. We coded the outcome variable as:

- outcome_encd = 1 for warning
- outcome_encd = 2 for citation
- outcome_encd = 3 for arrest

In the multinomial model, **warning** serves as the baseline category. The fitted model, therefore, estimates two logit equations:

1. log-odds of citation versus warning
2. log-odds of arrest versus warning

This formulation directly supports the interpretation of whether neighborhood wealth is associated with receiving a more punitive outcome relative to a warning.

The main explanatory variable is a constructed neighborhood wealth proxy, wealth index (see 2.2). We centered the wealth variable around its sample mean:

$$wealth_c = log(median\ sale\ price) - mean(log(median\ sale\ price))$$

Centering was used for two reasons. First, it improves the interpretability of main effects in the presence of interactions, since coefficients are then interpreted at an average level of neighborhood wealth rather than at an implausible zero value. Second, it helps reduce nonessential collinearity between lower-order and interaction terms.

We also included demographic covariates:

- subject race (subject_race_encd)

- subject sex (subject_sex_encd)
- subject age (subject_age)

The dataset encoding used in the model is:

- race: White = 1, Black = 2, Hispanic = 3, Asian/Pacific Islander = 4, Unknown = 5, Other = 6
- sex: Male = 1, Female = 2

A key goal of the project is to assess whether the effect of neighborhood wealth differs across demographic groups. To capture this heterogeneity, we included the following interaction terms:

- wealth × race
- wealth × sex

These interactions allow the wealth-outcome association to vary across racial categories, between males and females. This is important because a single global wealth coefficient would assume that neighborhood context affects all groups identically, which is not plausible given the project's motivating question.

The final model specification, therefore, included:

- main effect of centered wealth
- main effect of age
- categorical effects for race and sex
- wealth × race interactions
- wealth × sex interactions

$$log(\frac{P(Y=k)}{P(Y=Warning)}) = B_0^{(k)} + B_1^{(k)} * wealth + B_2^{(k)} * age + B_3^{(k)} * race + B_4^{(k)} * sex + \gamma^{(k)} * (wealth \times race) + \delta^{(k)} * (wealth\, x\, sex)$$

During assumption checking, age showed evidence of nonlinearity in the log-odds. To address this, we modeled age using a spline basis rather than a strictly linear term in the final multinomial specification. A spline is a flexible way to model nonlinear relationships by breaking a continuous variable (like age) into segments and fitting piecewise polynomial functions that are smoothly connected at predefined points called knots. Wealth, after log transformation, did not show the same level of nonlinearity in diagnostics and was therefore retained as a linear term. This decision preserves the interpretability of the neighborhood wealth effect while allowing age to enter the model flexibly.

In addition to coefficient-based inference, we generated predicted probabilities and marginal-effect visualizations. Building in multiple interactions makes interpretation complicated. The plots allowed us to translate the fitted model into substantively meaningful patterns, such as how the probability of warning, citation, or arrest changes as neighborhood wealth increases for different demographic profiles.

3.2. Model Training and Tuning

Although the project's primary goal is inferential, we still evaluated the model's out-of-sample performance to determine whether the fitted associations were at least reasonably stable beyond the estimation sample. This is important because a model that fits the training data well but collapses on new data may still be of limited value.

We used a temporal train-test split rather than a random split.

- Training set: stops from 2013-2015
- Test set: stops from 2016

A random split would mix observations across all years into both the training and testing sets, leading to more optimistic performance estimates due to temporal and spatial similarity between training and test observations. Since our wealth proxy is spatially structured and neighborhoods recur across years, a random split could allow the model to benefit from seeing very similar contexts in both sets. A year-based split provides a stricter, more realistic evaluation by asking whether relationships learned in earlier years carry forward to later stops.

The model training process involved two related estimation stages.

First, for inference, we fit an MLR using *statsmodels*. Because full-sample multinomial estimation with interactions and spline terms is computationally expensive on a dataset of this size, we drew a large stratified sample from the training data for the inferential fit. Stratification by outcome class ensured that all stop outcomes were represented proportionally in the estimation sample. This inferential model produced:

- Coefficient estimates
- Cluster-robust standard errors
- P-Values
- Z-Values
- Odds ratios

These quantities are the basis for the formal interpretation of the fitted associations.

Second, for held-out validation, we fit the same specification family to the full training set using a multinomial logistic regression implementation from scikit-learn, then evaluated it on the 2016 test set. This stage was not used for statistical inference; rather, it was used to assess whether the model produced sensible out-of-sample classification behavior. Our model produced test accuracy, macro F1 score, balanced accuracy, and other metrics. However, we chose not to report these values as they are not relevant to inferential goals.

Tuning in this project focused less on hyperparameter optimization and more on

specification tuning. The following were the key specification choices:

1. Whether to model age linearly or nonlinearly
2. If wealth required a nonlinear form after log transformation
3. Should interactions between wealth and demographic variables be included

These choices were guided by diagnostics and logic rather than predictive score optimization. This is consistent with the project's inferential orientation: the goal was not to maximize test-set accuracy at all costs, but to choose a model that best reflects the question being asked and the structure of the data.

3.3. Modeling Assumptions

Independence of Observations:
The model assumes that each observation (traffic stop) is independent of all others. In practice, this assumption is likely violated due to spatial and temporal clustering. Stops occurring in the same geographic area or within the same time period may be influenced by similar policing practices, neighborhood characteristics, or enforcement intensity.

To address this, we implemented cluster-robust standard errors, grouping observations by a combination of year and rounded geographic coordinates (latitude/longitude grid cells). This adjustment relaxes the strict independence assumption by allowing for correlation within clusters while maintaining independence across clusters. Although this does not fully eliminate dependence, it provides more reliable inference than naive standard errors.

Nominal Dependent Variable Categories:
Multinomial logistic regression assumes that the dependent variable consists of mutually exclusive categories with no inherent ordering. In this analysis, the outcome variable (warning, citation, arrest) satisfies these conditions: each stop is assigned exactly one outcome, and all modeled outcomes are distinct.

While one could argue that these outcomes have a conceptual severity ordering, we deliberately treated them as nominal rather than ordinal. Given the substantive goal of allowing effects (especially of wealth) to differ between citation and arrest, the nominal specification is more flexible and appropriate.

No Severe Multicollinearity:
The model assumes that predictor variables are not highly collinear. Severe multicollinearity can inflate standard errors, destabilize coefficient estimates, and make interpretation unreliable.

We assessed multicollinearity using Variance Inflation Factors (VIFs) computed on the design matrix. The maximum observed VIF was approximately 2.7, which is well below commonly used

thresholds (e.g., 5 or 10), indicating that multicollinearity is not severe. Additionally, we centered continuous variables such as age and wealth index to reduce nonessential collinearity introduced by interaction terms. Overall, multicollinearity does not appear to be a major concern in this model.

Term	VIF
wealth_c	2.6749
Intercept	2.5778
wealth_c:C(subject_race_encd)[T.2]	1.9127
wealth_c:C(subject_sex_encd)[T.2]	1.7662
C(subject_race_encd)[T.2]	1.1663
wealth_c:C(subject_race_encd)[T.3]	1.1320
C(subject_race_encd)[T.3]	1.1235
wealth_c:C(subject_race_encd)[T.4]	1.0343
wealth_c:C(subject_race_encd)[T.5]	1.0271
C(subject_race_encd)[T.4]	1.0242
C(subject_race_encd)[T.5]	1.0195
C(subject_sex_encd)[T.2]	1.0138
age_c	1.0097
wealth_c:C(subject_race_encd)[T.6]	1.0093
C(subject_race_encd)[T.6]	1.0090

Linearity Between Continuous Predictors and Log-Odds:
Multinomial logistic regression assumes that continuous predictors have a linear relationship with the log-odds of each outcome, unless otherwise specified. Violations of this assumption can lead to biased or misspecified effects.

We evaluated this assumption using diagnostic tests based on binary logistic regressions (citation vs. warning, arrest vs. warning) and Box–Tidwell-style interaction terms. These terms are used to test the linearity assumption between the continuous independent variables and dependent variables used in regression. These diagnostics indicated that the linearity assumption was violated for age but reasonably satisfied for the log-transformed wealth index.

Comparison	Wealth log p-value	Age log p-value
Citation vs Warning	0.8636	9.68e-07
Arrest vs Warning	0.9549	0.8817

To address this, we modeled age using a spline transformation, allowing for a flexible, nonlinear relationship between age and the log-odds of each outcome. The wealth index, after log transformation and centering, was retained as a linear term to preserve interpretability.

IIA (Independence of Irrelevant Alternatives):
A key assumption of MLR is the Independence of Irrelevant Alternatives (IIA). This assumption states that the relative odds between any two outcome categories are independent of the presence or

characteristics of other categories.

We assessed IIA using a sensitivity analysis, comparing model coefficients from the full multinomial model to those obtained after removing one outcome category and re-estimating the model. The results showed noticeable shifts in some coefficients, suggesting that the IIA assumption may not hold perfectly in this context.

This is not unexpected, as traffic stop outcomes may arise from related decision processes (e.g., escalation from warning to citation to arrest), which can violate the independence implied by IIA. As a result, the MLR model should be interpreted as a flexible associational model rather than a full representation of the decision process. The potential violation of IIA is acknowledged as a limitation of the modeling approach.

Assumption	Result	Interpretation
Zero-count race × sex cells	0	No issue
Maximum VIF	2.6749	Acceptable multicollinearity
Age nonlinearity	Yes	Spline recommended
Wealth nonlinearity	No	Linear OK
IIA relative change (wealth)	1.5868	Check sensitivity
IIA flag	Yes	Caution

4. Results

4.1. Results

Multinomial logit Coefficients: Refer to Figure 5 in the appendix.

Multinomial logit Odds-Ratios (Figure 6):

	Citation vs Warning	Arrest vs Warning
Intercept	0.2184	0.5522
Black	0.9962	2.0836
Hispanic	2.2225	2.8403
Asian/Pacific Islander	0.8106	0.4861
Unknown	1.3497	0.2609
Female	0.8743	0.4141
Wealth Index	0.8842	0.9214
Wealth Index x Black	1.1739	1.0709

Multinomial logit P-Values (Figure 7):

	Citation vs Warning	Arrest vs Warning
Intercept	9.16E-13	2.57E-01
Black	8.94E-01	5.11E-21
Hispanic	6.44E-58	5.87E-20
Asian/Pacific Islander	1.67E-02	1.29E-01
Unknown	8.53E-04	1.98E-02
Female	8.22E-10	4.63E-36
Wealth Index	6.77E-02	4.60E-01
Wealth Index x Black	1.14E-02	5.99E-01

Multinomial logit Z-Values (Figure 8):

	Citation vs Warning	Arrest vs Warning
Intercept	-7.1426	-1.1341
Black	-0.1337	9.4068
Hispanic	16.0426	9.1466
Asian/Pacific Islander	-2.3937	-1.5190
Unknown	3.3351	-2.3294
Other	-0.5544	0.0554
Female	-6.1406	-12.5380
Wealth Index	-1.8269	-0.7393
Wealth Index x Black	2.5314	0.5252

The inferential results from the MLR indicate that demographic factors, particularly race and sex, are the strongest predictors of traffic stop outcomes. The effect of neighborhood wealth is more nuanced and operates through interactions. Relative to warnings, several racial groups exhibit higher odds of more severe outcomes. For example, Hispanic individuals have over 2x higher odds of receiving a citation and nearly 3x higher odds of arrest, while Black individuals have more than 2x higher odds of arrest, all highly statistically significant ($p<.001$). In contrast, females have significantly lower odds of both citations (≈13% lower) and arrests (≈59% lower) compared to males, indicating a gender-based difference in enforcement severity. These effects are supported by small p-values and large z-values, suggesting they are statistically robust and significant.

While the main effect of neighborhood wealth is not statistically significant, interaction terms reveal that wealth influences outcomes differently across groups. In particular, because the wealth index is defined as the logarithm of the median housing price, a one-unit increase corresponds to approximately a 2.7x increase in price. Interpreted on a more realistic scale, a 10% increase in housing price is associated with approximately a

1-2% increase in the odds of citation for Black individuals, indicating a modest but statistically significant effect. Overall, these results suggest that traffic stop outcomes are strongly associated with demographic characteristics, and that the effect of neighborhood wealth depends on the group being observed, rather than acting uniformly across the population.

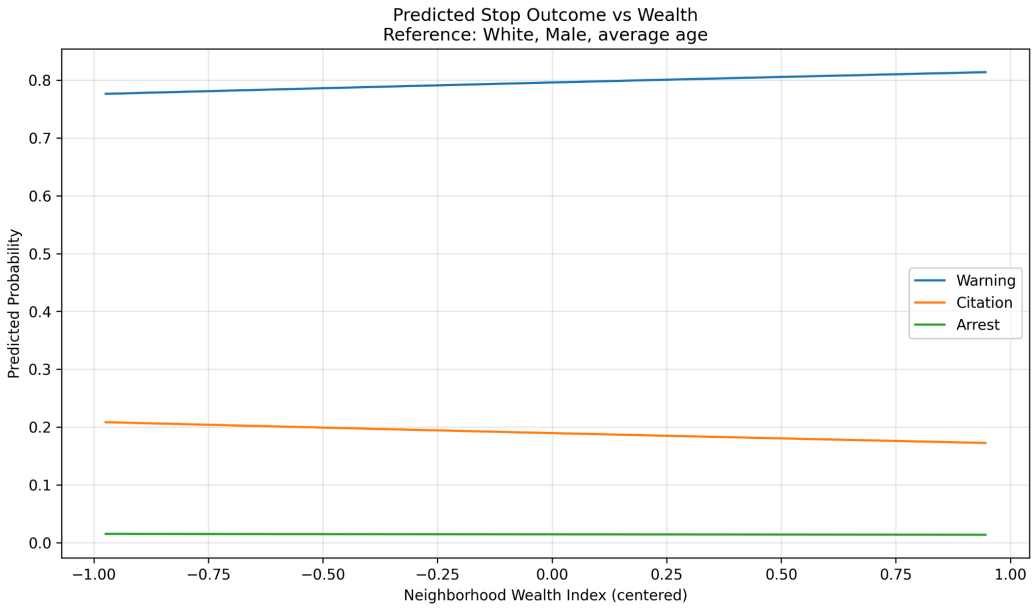

Figure 9. Predicted Stop Outcome vs Wealth

This figure displays the predicted probability of each stop outcome as neighborhood wealth increases, holding race fixed at White, sex fixed at male, and age at the sample average. The results indicate a modest but systematic relationship between neighborhood wealth and stop outcomes. Specifically, as wealth increases, the predicted probability of receiving a warning rises slightly, while the probability of receiving a citation declines. However, the probability of arrest remains very low and shows little variation across the wealth distribution.

These patterns suggest that drivers stopped in wealthier areas are somewhat more likely to experience less severe stop outcomes, primarily from citations to warnings. However, the magnitude of these changes is relatively small, indicating that while neighborhood wealth appears to influence stop outcomes, it is not a dominant determinant.

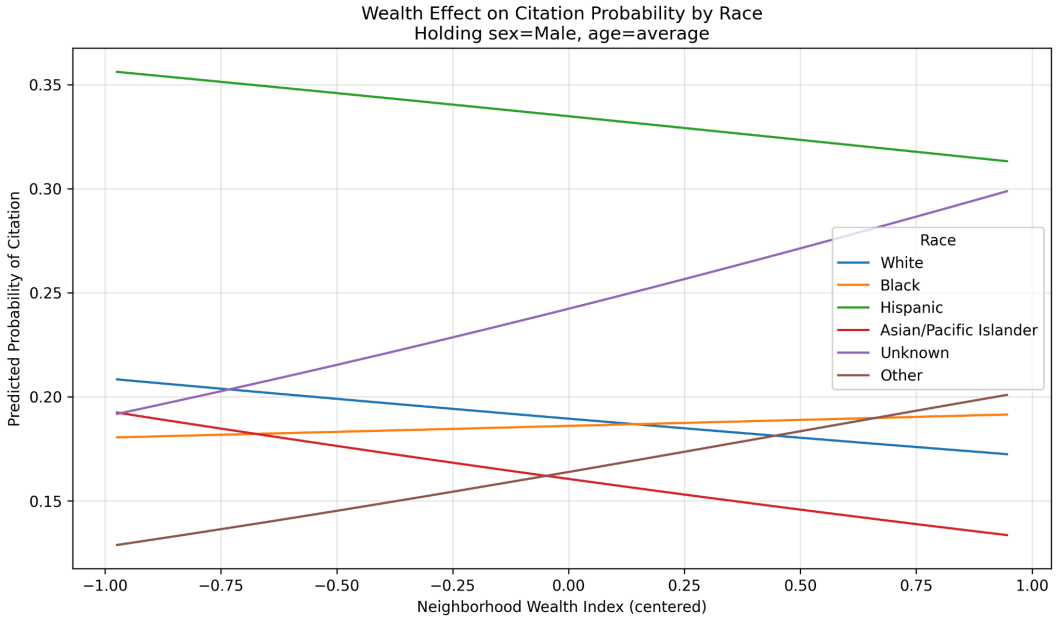

Figure 10. Wealth Effect on Citation Probability by Race

This figure shows how the relationship between neighborhood wealth and the probability of receiving a citation varies across racial groups, holding sex constant at male and age at the sample average. The results reveal large heterogeneity in the wealth effect. For White and Hispanic drivers, the probability of receiving a citation declines as neighborhood wealth increases, suggesting that stops in wealthier areas are associated with more lenient outcomes for these groups. In contrast, the relationship for Black drivers appears slightly positive, indicating that citation probability may increase modestly as neighborhood wealth rises. Other racial categories exhibit mixed patterns, with some groups showing increasing citation probabilities in wealthier neighborhoods.

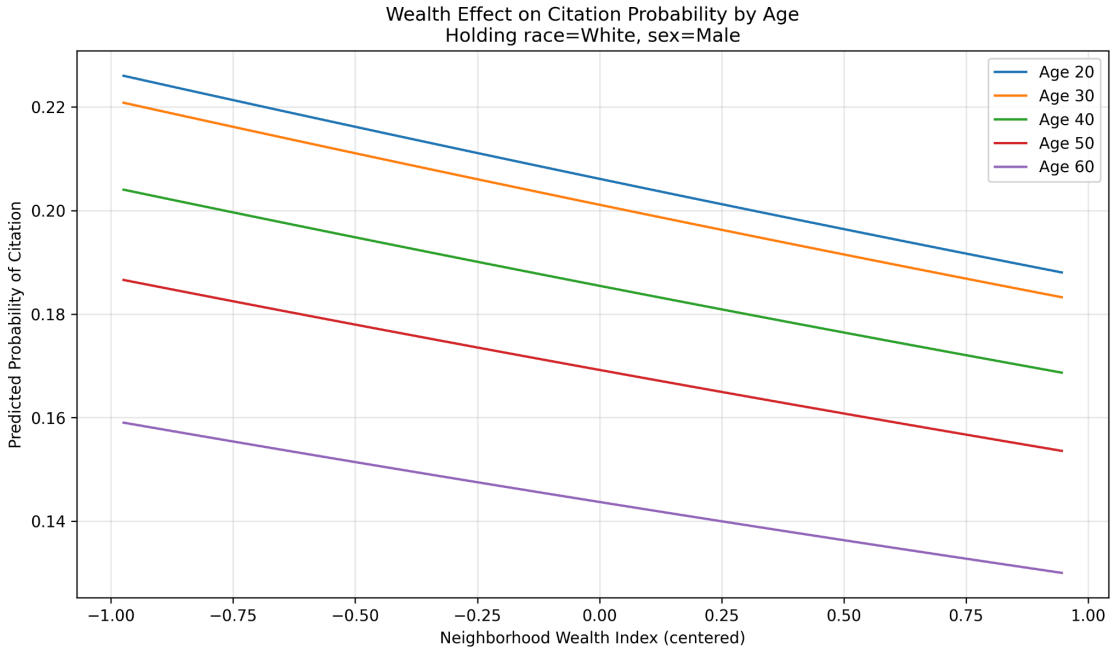

Figure 11. Wealth Effect on Citation Probability by Age

The figure above examines whether the effect of neighborhood wealth on citation probability differs across age groups, holding race fixed at White and sex fixed at male. The results show that younger drivers have consistently higher predicted probabilities of receiving citations than older drivers across all levels of neighborhood wealth. However, the slope of the relationship between wealth and citation probability is similar across age groups, with citation likelihood declining gradually as wealth increases for drivers of all ages.

4.2 Model Performance

While we evaluate model performance using standard metrics, our primary goal is inference rather than prediction. The dataset exhibits substantial class imbalance, with warnings accounting for the majority of outcomes and arrests occurring relatively infrequently. As a result, traditional accuracy metrics can be misleading, since a model that simply predicts the majority class would achieve high accuracy without providing meaningful insight.

In practice, this class imbalance was reflected in the model's predictions on the test data. The model tended to predict the majority class (warnings) much more often than the minority classes (citations or arrests), and the minority classes were difficult for the model to classify accurately. While this would be a major concern in a purely predictive project, it is less critical in this analysis because our primary goal is inference. The purpose of the model is to estimate and interpret the relationships between neighborhood wealth, demographic characteristics, and stop outcomes, rather than to maximize classification accuracy. Therefore, the model is still useful for understanding associations in the data, even if its predictive performance for rare outcomes is limited.

5. Discussion and Conclusion

5.1. Discussion of Impact

The results described in the above sections indicate that while wealth index alone is not strongly associated with the stop outcome, its interactions with certain demographic characteristics are. Our results tell us that there is a statistically significant association between both race and sex with stop outcomes. Specifically, the odds of citation decrease as wealth index increases for White and Hispanic individuals, while the odds of citation increase slightly as wealth index increases for Black individuals. Additionally, we found that a 10% increase in wealth index is associated with a 1-2% increase in the odds of citation for Black individuals.

Combining these results, we observe a pattern: an increase in an area's general wealth is

associated with an increase in the odds that a Black individual receives a citation there. Although our model is not entirely comprehensive of all possible confounders, these results indicate a potential systemic issue in the Nashville Police Department. Because our model's results did not find this trend in White or Hispanic individuals, we have an indication that this is disproportionately impacting the Black community in Nashville. Evidence such as this serves as a significant step in the right direction towards addressing the issue of racial bias in the Nashville Police Department. From here, these findings could be used for a multitude of purposes, such as further research, community activism, or policy implementation.

One of the main project challenges we faced was the lack of geospatial data in our supplemental dataset. To combat this, we used OpenStreetMap's API to geocode individual street addresses into latitude and longitude values. From here, we constructed a wealth index variable to quantify the general wealth of different neighborhoods in Nashville. These two solutions allowed us to begin building our baseline model and served as the first steps towards addressing our project objective.

Our model revealed results that our initial EDA missed. This project serves as a reminder to the field of data science of the importance of rigorous modeling with intentionally specified interaction terms, even when EDA yields significant preliminary results.

5.2. Limitations and Future Work

Some limitations of our project include our model failing to meet the necessary assumptions and the potential for simultaneity among our variables.

One assumption of the MLR model is Independence of Irrelevant Alternatives (IIA). For the IIA condition to be met in our model, the probability of an individual receiving a specific stop outcome over another must be unaffected by the addition or removal of another stop outcome category. However, we cannot confidently say that our model meets this assumption, as the police department likely decides which type of stop outcome to assign to an individual based on the knowledge that stop outcomes are related rather than independent categories. For example, if warnings were removed as a stop outcome, the probability that an individual would receive a citation rather than an arrest would likely increase. These factors indicate that the IIA assumption might not hold for our model.

To concretely determine whether IIA holds, we could run a sensitivity model in the future. This would tell us whether there is a significant shift in the coefficients, indicating that our model does not meet the IIA assumption. Although IIA is an important assumption of our model, our project will continue to use MLR. Because the goal of our project is inference rather than prediction, this is the industry standard. However, we do acknowledge that this is a limitation of our project.

Another limitation of our project is the potential simultaneity of the wealth index and stop outcome

variables. Simultaneity occurs when two variables affect each other simultaneously rather than when one variable affects the other. This causes the explanatory variable to become correlated with the error term. When simultaneity occurs, it can lead to endogeneity (biased coefficient estimates). In our project, we are potentially concerned with the simultaneity between the wealth index and stop-outcome variables. Our project was designed to investigate the relationship between wealth index and stop outcome. However, we are concerned that the wealth index of an area might be affected by the policing patterns in that area. For example, a specific stop outcome category being highly prevalent in a specific area could potentially have impacted house prices in that area. This, in turn, would have impacted the creation of the wealth index variable.

To determine the presence of simultaneity in our project, we could run an Instrumental Variable model, such as Two-Stage Least Squares, because these models are designed to account for endogeneity. We would then compare the results from our MLR model and IV model. If there was a significant difference in coefficient estimates between the two models, then we know there is simultaneity between our variables. However, it might be challenging to find an IV that works for our data. To mitigate simultaneity issues, we could reconstruct our wealth index variable using housing prices from before our SOPP data timeframe. Although simultaneity is a real concern, we expect that future work will show it has a minuscule effect on our project.

In the future, we plan to improve the interpretability of our wealth index variable, adapt the validation strategy for our MLR model, adjust our baseline model, and run a sensitivity analysis.

We originally log-transformed the wealth index variable to account for outliers in our housing data. To make this variable more interpretable to the general population, we plan to transform the wealth index variable back to its original scale and bin it into categories based on standard deviation units. This will allow us to explain the effects of the wealth index variable in terms of high or low wealth rather than in complex decimals.

Currently, we use a temporal train-test split for our data. This is working well for our model, but in the future, we plan to adapt the validation strategy by implementing a spatial cross-validation technique. This approach is more complex than our current split, but it is also more rigorous as it trains and tests the model on different geographic regions. This ensures the separation of nearby data points, thereby increasing the model's overall robustness.

Our initial baseline MLR model identifies the relationship between wealth index and stop outcome by accounting for race, age, and sex, and their interactions with wealth index. However, it is difficult to interpret the individual effect of wealth index on stop outcome. In the future, we plan to run multiple levels of our baseline model, increasing the complexity at each level. We will begin with our model controlling for age, race, and sex. Next, we will run our model controlling for wealth index, age, race, and sex. Then, we will run our original baseline model with interactions among age, race, sex, and wealth index. This will allow us to compare the results from each model and determine whether wealth index effects impact stop outcomes differently across individuals of

different demographics.

In the future, we plan to run a sensitivity analysis to assess the decision to potentially remove the unknown race category from our model. We are considering removing the unknown category because we are concerned it indicates a quality issue in our data. A sensitivity model would show whether there was a significant shift in coefficients, indicating that removing the unknown race category had unexpected effects on our project that we need to address.

Additionally, we are considering expanding our model to our entire dataset. Currently, our MLR model is using only 75,000 observations out of the ~1,300,000 total. We made this choice because fitting the model to a dataset as large as ours is very computationally expensive. However, expanding the model will increase the statistical power, so we are currently developing a plan to implement this.

The main ethical concern of our project is the inadvertent reinforcement of certain negative stereotypes about individuals of specific demographics. It is not our intention to suggest that certain groups of individuals *deserve* harsher stop outcomes, such as an arrest instead of a warning or vice versa. Rather than investigating which demographic characteristics are associated with different stop outcomes, our project aims to strictly examine the relationship between stop outcomes and neighborhood wealth by controlling for demographic factors. To combat this potential misinterpretation of our objective, we will continue to emphasize the goal of this project, which is to determine associations rather than causal relationships.

6. Reproducibility

6.1. Data Access

Our main dataset can be found on the Stanford Open Policing Project Data website, or at this link. Once on the website, you can scroll down to Tennessee and download the raw file titled Nashville to access our main dataset.

Our supplemental dataset was obtained from Kaggle at this link. This dataset is licensed under a CCO: Public Domain license, meaning it is publicly available for analysis, modification, and adaptation without explicit permission from the publisher.

6.2. GitHub Repository

Our project's GitHub repository can be found here. To clone the repository, users can run the following command in their terminal:

```
git clone https://github.com/utexas-SDS-357/sds357-project-sp26-hyperparameter-tuners-1.git
```

To navigate to the project directory, users can run the following command in their terminal:

cd sds357-project-sp26-hyperparameter-tuners-1

To reproduce our geocoding process, users can run the geocoding.py file located in the geocoding branch. Before running this code, users will need to install the following packages: certifi, pandas, tqdm, and geopy.

To reproduce our general analysis, we are working to provide access to the .ipynb notebooks and associated files through our GitHub repository. These materials will include the code used for data cleaning, variable construction, model estimation, and figure generation. Users will be able to download and run the notebooks locally to replicate the results provided in our analysis.

7. References

Stanford Open Policing Project. (n.d.). Traffic stop data: Nashville, TN. Stanford University. https://openpolicing.stanford.edu/data/

tmthyjames. (n.d.). *Nashville housing data*. Kaggle. https://www.kaggle.com/datasets/tmthyjames/nashville-housing-data

8. Appendix

Multinomial logit Coefficients (Figure 5)

	Citation vs Warning	Arrest vs Warning
Intercept	-1.5217	-0.5938
C(subject_race_encd)[Black]	-0.0038	0.7341
C(subject_race_encd)[Hispanic]	0.7986	1.0439
C(subject_race_encd)[Asian/Pacific Islander]	-0.2100	-0.7213
C(subject_race_encd)[Unknown]	0.2999	-1.3437
C(subject_race_encd)[Other]	-0.1754	0.0500
C(subject_sex_encd)[Female]	-0.1343	-0.8816
wealth_c	-0.1231	-0.0818
wealth_c:C(subject_race_encd)[Black]	0.1604	0.0685
wealth_c:C(subject_race_encd)[Hispanic]	0.0021	-0.3265
wealth_c:C(subject_race_encd)[Asian/Pacific Islander]	-0.1084	-0.4386

wealth_c:C(subject_race_encd)[Unknown]	0.4318	0.8636
wealth_c:C(subject_race_encd)[Other]	0.4255	1.2525
wealth_c:C(subject_sex_encd)[Female]	0.0229	-0.1324
bs(age_c, df=4, include_intercept=False)[0]	0.3107	-3.8910
bs(age_c, df=4, include_intercept=False)[1]	-0.0970	-2.2290
bs(age_c, df=4, include_intercept=False)[2]	-0.1282	-6.2966
bs(age_c, df=4, include_intercept=False)[3]	-3.1252	-7.3766

Multinomial logit Odds-Ratios (Figure 6)

	Citation vs Warning	Arrest vs Warning
Intercept	0.2184	0.5522
C(subject_race_encd)[Black]	0.9962	2.0836
C(subject_race_encd)[Hispanic]	2.2225	2.8403
C(subject_race_encd)[Asian/Pacific Islander]	0.8106	0.4861
C(subject_race_encd)[Unknown]	1.3497	0.2609
C(subject_race_encd)[Other]	0.8391	1.0513
C(subject_sex_encd)[Female]	0.8743	0.4141
wealth_c	0.8842	0.9214
wealth_c:C(subject_race_encd)[Black]	1.1739	1.0709
wealth_c:C(subject_race_encd)[Hispanic]	1.0021	0.7215
wealth_c:C(subject_race_encd)[Asian/Pacific Islander]	0.8973	0.6449
wealth_c:C(subject_race_encd)[Unknown]	1.5400	2.3718
wealth_c:C(subject_race_encd)[Other]	1.5303	3.4990
wealth_c:C(subject_sex_encd)[Female]	1.0231	0.8760
bs(age_c, df=4, include_intercept=False)[0]	1.3644	0.0204
bs(age_c, df=4, include_intercept=False)[1]	0.9076	0.1076
bs(age_c, df=4, include_intercept=False)[2]	0.8797	0.0018
bs(age_c, df=4, include_intercept=False)[3]	0.0439	0.0006

Multinomial logit P-Values (Figure 7)

	Citation vs Warning	Arrest vs Warning
Intercept	9.16E-13	2.57E-01
C(subject_race_encd)[Black]	8.94E-01	5.11E-21
C(subject_race_encd)[Hispanic]	6.44E-58	5.87E-20
C(subject_race_encd)[Asian/Pacific Islander]	1.67E-02	1.29E-01
C(subject_race_encd)[Unknown]	8.53E-04	1.98E-02

C(subject_race_encd)[Other]	5.79E-01	9.56E-01
C(subject_sex_encd)[Female]	8.22E-10	4.63E-36
wealth_c	6.77E-02	4.60E-01
wealth_c:C(subject_race_encd)[Black]	1.14E-02	5.99E-01
wealth_c:C(subject_race_encd)[Hispanic]	9.85E-01	1.93E-01
wealth_c:C(subject_race_encd)[Asian/Pacific Islander]	5.65E-01	6.17E-01
wealth_c:C(subject_race_encd)[Unknown]	4.85E-02	7.33E-02
wealth_c:C(subject_race_encd)[Other]	4.92E-01	1.74E-02
wealth_c:C(subject_sex_encd)[Female]	5.75E-01	2.99E-01
bs(age_c, df=4, include_intercept=False)[0]	2.43E-01	4.72E-08
bs(age_c, df=4, include_intercept=False)[1]	5.25E-01	1.70E-09
bs(age_c, df=4, include_intercept=False)[2]	7.39E-01	2.04E-07
bs(age_c, df=4, include_intercept=False)[3]	1.18E-08	8.39E-04

Multinomial logit Z-Values (Figure 8)

	Citation vs Warning	Arrest vs Warning
Intercept	-7.1426	-1.1341
C(subject_race_encd)[Black]	-0.1337	9.4068
C(subject_race_encd)[Hispanic]	16.0426	9.1466
C(subject_race_encd)[Asian/Pacific Islander]	-2.3937	-1.5190
C(subject_race_encd)[Unknown]	3.3351	-2.3294
C(subject_race_encd)[Other]	-0.5544	0.0554
C(subject_sex_encd)[Female]	-6.1406	-12.5380
wealth_c	-1.8269	-0.7393
wealth_c:C(subject_race_encd)[Black]	2.5314	0.5252
wealth_c:C(subject_race_encd)[Hispanic]	0.0190	-1.3024
wealth_c:C(subject_race_encd)[Asian/Pacific Islander]	-0.5756	-0.5008
wealth_c:C(subject_race_encd)[Unknown]	1.9732	1.7909
wealth_c:C(subject_race_encd)[Other]	0.6875	2.3785
wealth_c:C(subject_sex_encd)[Female]	0.5611	-1.0377
bs(age_c, df=4, include_intercept=False)[0]	1.1667	-5.4616
bs(age_c, df=4, include_intercept=False)[1]	-0.6355	-6.0245
bs(age_c, df=4, include_intercept=False)[2]	-0.3331	-5.1953
bs(age_c, df=4, include_intercept=False)[3]	-5.7030	-3.3394